

“Exploratory Data Analysis of Netflix Movies and TV Shows Dataset”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies

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FOR THE INTERNSHIP

IN

DATA SCIENCE

By

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Introduction

The rapid growth of digital streaming platforms has significantly transformed the way audiences consume entertainment content. Among these platforms, Netflix has emerged as one of the leading providers of movies and television shows worldwide, offering a diverse range of content across various genres, languages and regions.

This project focuses on performing Exploratory Data Analysis (EDA) on the Netflix dataset to gain meaningful insights into the platform's content distribution and trends. The dataset contains information about movies and TV shows, including attributes such as title, type, release year, genre, duration, country.

The primary objective of this analysis is to understand the composition and evolution of Netflix's content library. By analyzing key factors such as the distribution of movies and TV shows, trends in content production over the years and the popularity of different genres, this study aims to uncover patterns that reflect the growth and strategy of the platform.

Data preprocessing techniques were applied to handle missing values and ensure data consistency. Various visualization methods, including pie charts, bar graphs, histograms and area plots, were used to present the findings in a clear and interpretable manner.

Overall, this project provides a comprehensive overview of Netflix's content trends and demonstrates how data analysis and visualization can be used to extract valuable insights from real world datasets.

Objectives

- ❖ To load and explore the Netflix dataset using Pandas to understand its structure and features.
- ❖ To preprocess the dataset by handling missing values and converting data into appropriate formats for analysis.
- ❖ To analyze the distribution of content types (Movies vs TV Shows) available on Netflix.
- ❖ To examine the year-wise growth of Netflix content to identify patterns and trends over time.
- ❖ To identify the most popular genres and categories of content on the platform.
- ❖ To understand the growth pattern of Netflix content across different years.
- ❖ To highlight the major content-producing countries and understand their contribution to Netflix's library.

Methodology

This project follows a structured approach to perform Exploratory Data Analysis (EDA) on the Netflix dataset. The methodology is divided into the following key stages:

1. Data Collection

The dataset used in this project is the Netflix Movies and TV Shows dataset obtained from Kaggle. It contains detailed information about various titles available on Netflix, including attributes such as type, genre, release year, duration, country and date added.

2. Data Preprocessing

Data preprocessing was performed to ensure the dataset was clean and suitable for analysis. This involved handling missing values by replacing them with appropriate placeholders such as Unknown or removing rows with minimal missing data.

3. Data Transformation

The dataset was transformed to make it suitable for analysis. The date_added column was converted into datetime format, and the genre column was split into multiple values to identify the most common genres. The data was also grouped and aggregated based on different attributes such as year, type and country.

4. Exploratory Data Analysis (EDA)

It was conducted to uncover patterns and trends within the dataset. Key analyses included identifying the proportion of movies and TV shows, examining content growth over the years, studying popular genres, studying duration patterns and identifying top content producing countries.

5. Data Visualization

Various visualization techniques were used to present the findings effectively. Pie charts, bar graphs and area plots were created using Matplotlib to make the analysis more intuitive and visually appealing.

Data Analysis

- ❖ To load and explore the Netflix dataset using Pandas to understand its structure and features.

Code:

```
import pandas as pd
```

```
df = pd.read_csv("C:\\Users\\Dell\\Downloads\\netflix_titles.csv")
```

```
# Display first 5 rows
```

```
print("First 5 Rows:\n")
```

```
print(df.head())
```

```
# Display last 5 rows
```

```
print("\nLast 5 Rows:\n")
```

```
print(df.tail())
```

```
# Display dataset info
```

```
print(df.info())
```

Output:

Data columns (total 5 rows)

	Show id	type	title	director	cast	country
0	S1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood	United States
1	S2	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabiha Akkari, Sofia Lesaffre, Salim Kechiouche, Noureddine Farihi, Geert Van Rempelberg, Bakary Diombera	France

2	S3	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, Henry Thomas, Kristin Lehman, Samantha Sloyan, Igby Rigney, Rahul Kohli, Annarah Cymone, Annabeth Gish, Alex Essoe, Rahul Abburi, Matt Biedel, Michael Trucco, Crystal Balint, Louis Oliver	United States
3	S4	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, Sofia Carson, Liza Koshy, Ken Jeong, Elizabeth Perkins, Jane Krakowski, Michael McKean, Phil LaMarr	Canada
4	S5	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra Duah, Nick Medley, Mutabaruka, Afemo Omilami, Reggie Carter, Mzuri	United States, Ghana, Burkina Faso, United Kingdom, Germany, Ethiopia

	date_added	release_year	Rating	Duration	listed_in
0	25-Sep-21	2020	PG-13	90 min	Documentaries
1	24-Sep-21	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Action & Adventure
2	24-Sep-21	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries
3	24-Sep-21	2021	PG	91 min	Children & Family Movies
4	24-Sep-21	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies

	Description
0	As her father nears the end of his life, filmmaker Kirsten Johnson stages his death in inventive and comical ways to help them both face the inevitable.
1	To protect his family from a powerful drug lord, skilled thief Mehdi and his expert team of robbers are pulled into a violent and deadly turf war.
2	The arrival of a charismatic young priest brings glorious miracles, ominous mysteries and renewed religious fervor to a dying town desperate to believe.
3	Equestria's divided. But a bright-eyed hero believes Earth Ponies, Pegasi and Unicorns should be pals – and, hoof to heart, she’s determined to prove it.
4	On a photo shoot in Ghana, an American model slips back in time, becomes enslaved on a plantation and bears witness to the agony of her ancestral past.

Data columns (total 12 rows)

	Column	Non-Null Count	dtype
0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	object
6	date_added	8797 non-null	object
7	release_year	8807 non-null	int64
8	rating	8803 non-null	object
9	duration	8804 non-null	object
10	listed_in	8807 non-null	object
11	description	8807 non-null	object

Range Index: 8807 entries, 0 to 8806

dtypes: int64(1), object(11)

Memory usage: 825.8+ KB

- ❖ To preprocess the dataset by handling missing values and converting data into appropriate formats for analysis.

Code:

```
import pandas as pd

df = pd.read_csv("C:\\Users\\Dell\\Downloads\\netflix_titles.csv")

df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Unknown')

df = df.dropna(subset=['date_added', 'rating', 'duration'])
print(df.head())

df.isnull().sum()
```

Output:

show_id	0
type	0
title	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	0
duration	0
listed_in	0
description	0

dtype: int64

Interpretation

The dataset was cleaned by converting data into appropriate formats and handling missing values. The date column was properly formatted for analysis. Now the dataset is clean, structured and ready for further analysis.

❖ To analyze the distribution of content types (Movies vs TV Shows) available on Netflix

Code:

```
plt.figure()  
df['type'].value_counts().plot(kind='pie', autopct='%1.1f%%')  
plt.title("Distribution of Movies vs TV Shows")  
plt.ylabel("")  
plt.show()
```

Output:

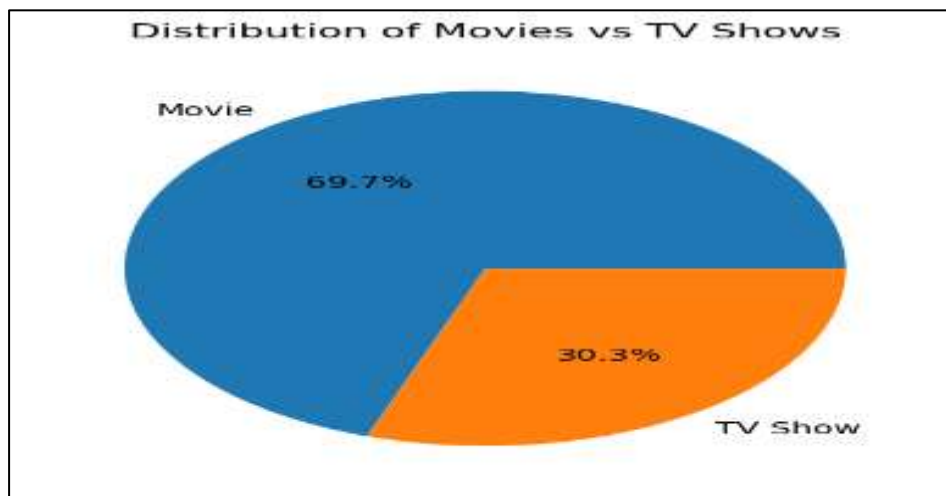


Fig.1

Interpretation

The pie chart shows that Movies make up 69.7% of the total content on Netflix, while TV Shows account for 30.3%. This indicates that movies are significantly more available on the platform compared to TV shows.

Conclusion

From the analysis, it can be concluded that Netflix's content library is dominated by movies. This suggests that the platform places greater emphasis on movie content, offering users a wider variety of films compared to TV shows.

- ❖ To examine the year-wise growth of Netflix content to identify patterns and trends over time.

Code:

```
df['release_year'].value_counts().sort_index().plot()  
plt.title("Content Growth Over Years")  
plt.show()
```

Output:

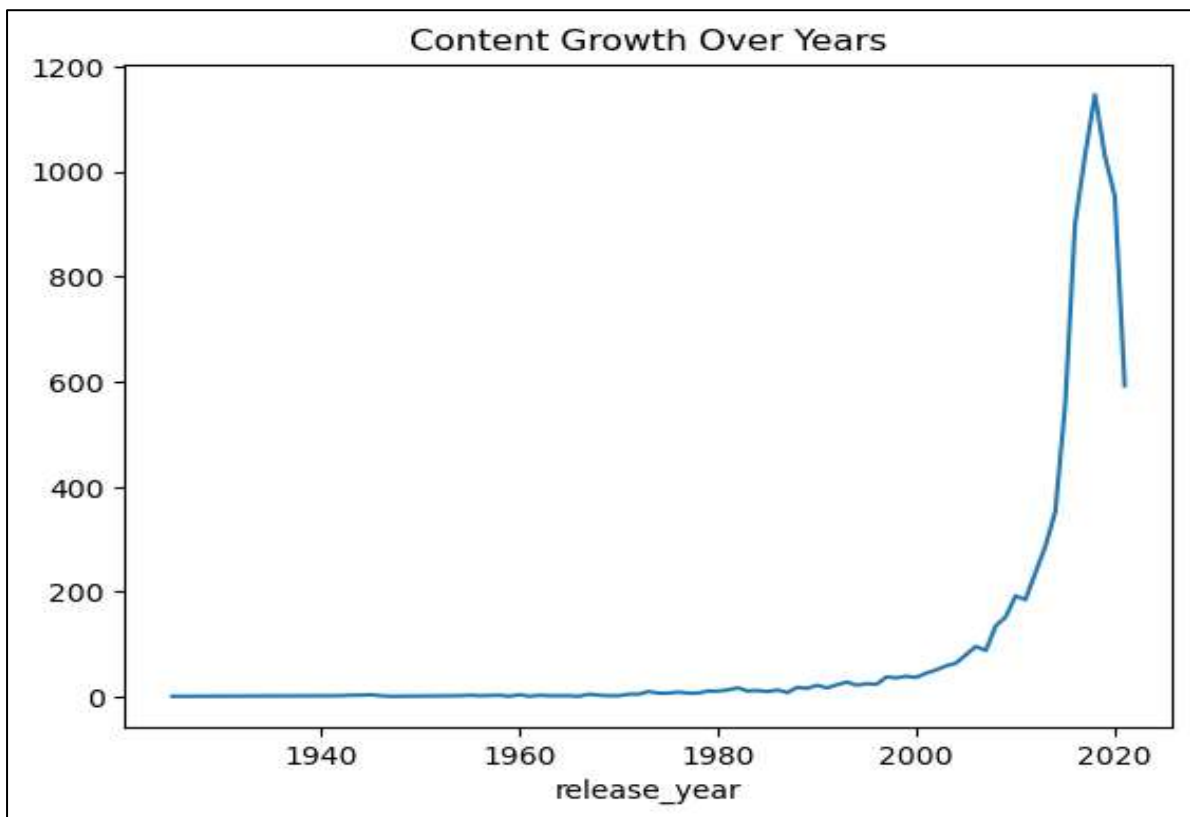


Fig.2

Interpretation

The graph shows the growth of Netflix content over the years. From the early years until around 2000, the number of releases remained very low and almost constant. After 2000, there is a gradual increase in content and from around 2015 onwards, there is a sharp rise in the number of movies and TV shows added. The content reaches its peak around 2019–2020, followed by a slight decline.

Conclusion

The analysis indicates that Netflix experienced rapid growth in content production in recent years, especially after 2015. This reflects the platform's expansion and increasing investment in content. The slight decline after the peak may indicate changes in content strategy or external factors affecting production.

❖ To identify the most popular genres and categories of content on the platform.

Code:

```
top_genres = df['listed_in'].str.split(' ', expand=True).stack().value_counts().head(10)
```

```
plt.figure()
```

```
top_genres.sort_values().plot(kind='barh')
```

```
plt.title("Top 10 Genres")
```

```
plt.xlabel("Count")
```

```
plt.show()
```

Output:

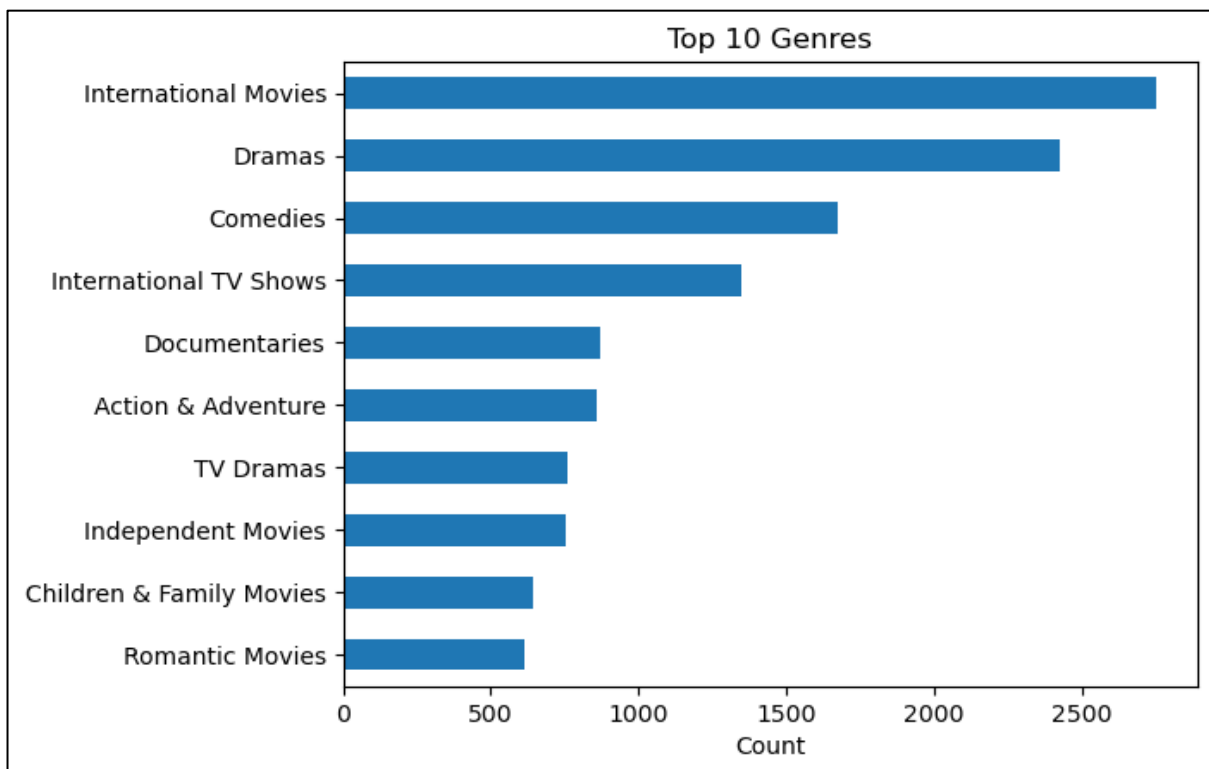


Fig.3

Interpretation

The bar chart shows the top 10 most common genres on Netflix. International Movies have the highest count, followed by Dramas and Comedies. International TV Shows and Documentaries are also among the frequently available genres. Other genres like Action & Adventure, TV Dramas and Independent Movies have moderate representation, while Children & Family Movies and Romantic Movies have comparatively lower counts.

Conclusion

From the analysis, it can be concluded that Netflix offers a wide variety of content, with a strong focus on International Movies, Dramas and Comedies. This indicates that the platform prioritizes globally diverse and popular entertainment genres to cater to a broad audience.

❖ To understand the growth pattern of Netflix content across different years.

Code:

```
top_years = df['release_year'].value_counts().head(10)
print(top_years)
```

```
plt.figure()
df['release_year'].plot(kind='hist', bins=20)
plt.title("Distribution of Release Years")
plt.xlabel("Year")
plt.show()
```

Output:

release_year	Count
2018	1146
2017	1030
2019	1030
2020	953
2016	901
2021	592
2015	555
2014	352
2013	286
2012	236

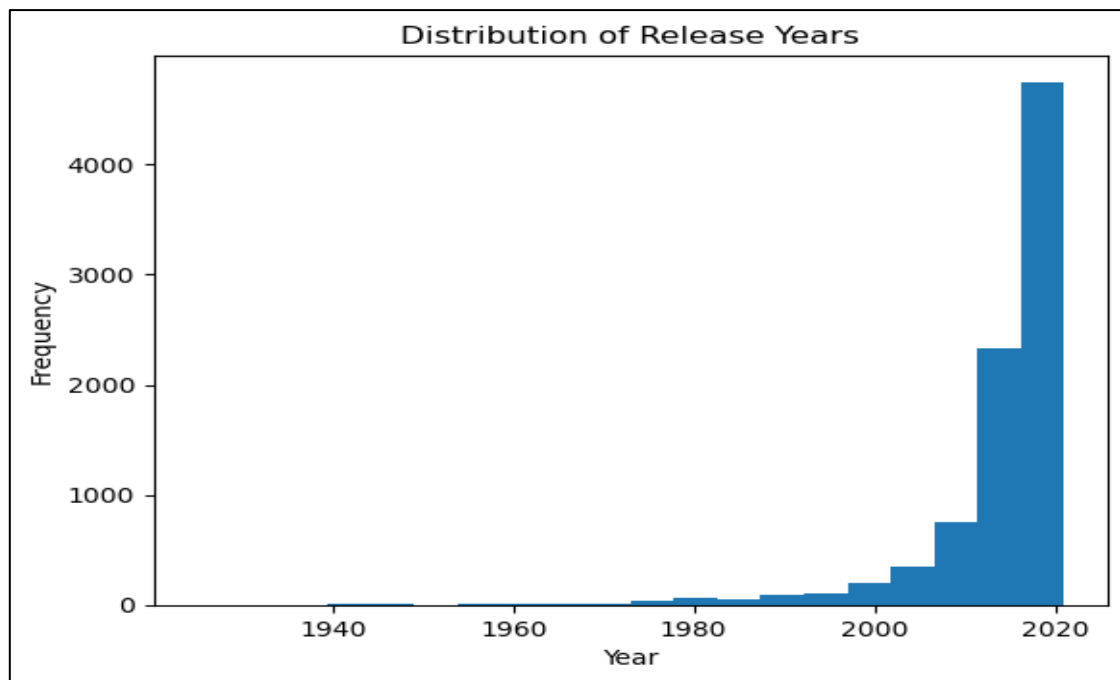


Fig.4

Interpretation

The histogram and data show that most of the content on Netflix is concentrated in recent years, especially after 2015. The highest number of releases occurred in 2018, followed by 2017 and 2019. Earlier years have very few releases, indicating low content availability in the past.

Conclusion

The analysis shows that Netflix has significantly increased its content production in recent years, with a peak around 2018–2020. This indicates rapid platform growth and higher investment in content during this period.

❖ **To highlight the major content-producing countries and understand their contribution to Netflix’s library.**

Code:

```
top_countries = df['country'].value_counts().head(10)
print(top_countries)
```

```
import matplotlib.pyplot as plt
```

```
plt.figure()
```

```
top_countries.plot(kind='bar')
```

```
plt.title("Top 10 Content Producing Countries")
plt.xlabel("Country")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```

Output:

Country	
United States	2809
India	972
United Kingdom	418
Japan	243
South Korea	199
Canada	181
Spain	145
France	124
Mexico	110
Egypt	106

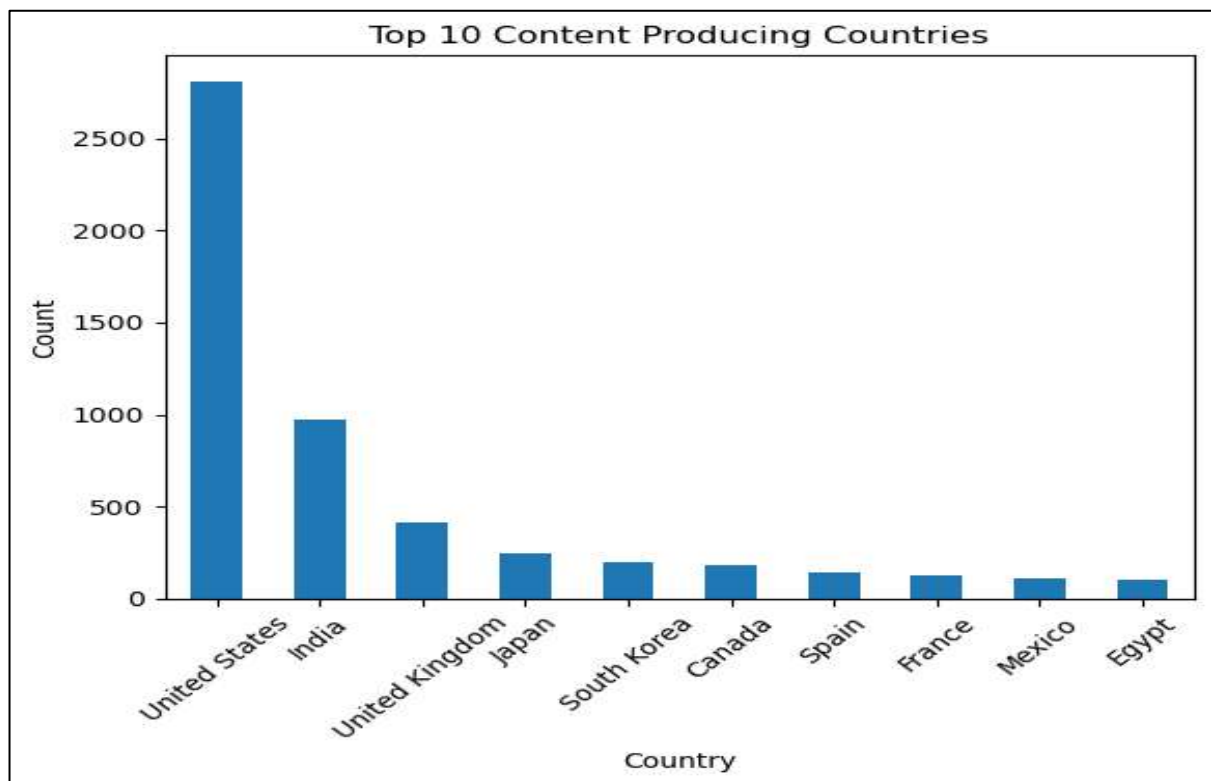


Fig.5

Interpretation

The bar graph shows the top countries contributing content to Netflix. The United States has the highest number of titles, followed by India. The United Kingdom, Japan, and South Korea also contribute a significant amount of content. Other countries such as Canada, Spain, France, and Mexico have comparatively lower contributions.

Conclusion

From the analysis, it can be concluded that Netflix content is mainly dominated by the United States, with India being the second-largest contributor. This indicates a strong presence of Netflix in these regions, while other countries contribute moderately to the overall content library.

Overall Conclusion

This project analyzed the Netflix dataset to understand content trends, genres and distribution across different factors. The analysis revealed that movies make up a larger portion of Netflix's content compared to TV shows, indicating a strong focus on film based entertainment.

Over the years, Netflix has shown significant growth in content production, especially after 2015, with a peak observed around 2018–2020. This reflects the platform's rapid expansion and increased investment in content.

In terms of genres, International Movies, Dramas and Comedies were found to be the most common, highlighting Netflix's focus on diverse and globally appealing content. Additionally, most TV shows consist of a single season and movies generally follow a standard duration range.

Country wise analysis showed that the United States and India are the leading contributors to Netflix's content library, indicating strong regional production and audience demand in these areas.

Overall, the project demonstrates how data analysis and visualization can effectively uncover meaningful insights from real world datasets, helping to better understand content strategies and trends in the streaming industry.